Extracting Product Announcements from SEC Form 8-K Filings

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1 Introduction

SEC Form 8-K filings disclose important corporate events, but the information is usually embedded in lengthy documents. This project uses a large language model (LLM) to identify product-related announcements and convert them into a table suitable for further analysis.

2 Data and Method

I collected 100 Form 8-K filings from selected publicly traded technology companies using SEC EDGAR. The company ticker JSON endpoint provided CIK identifiers, while the EDGAR Atom feed provided filing indexes. The main documents and relevant exhibits were downloaded and parsed with BeautifulSoup.

LLM-Based Extraction: Google Gemini 3.1 Flash Lite (`gemini-3.1-flash-lite`) was used to identify new product announcements from SEC Form 8-K filings. The model was prompted to return structured JSON records containing product names and descriptions under 180 characters. spaCy was used for named entity recognition (NER) to identify organizations in the filing text.

The extraction process followed four steps:

- **Retrieve:** Collect filing links and document text from EDGAR.
- **Clean:** Remove HTML markup and non-content elements.
- **Extract:** Use spaCy organization recognition and Gemini 3.1 Flash Lite to identify product mentions and return JSON.
- **Save:** Write company, ticker, filing time, product name, and description to CSV in Google Drive.

The prompt requested only newly announced products or services, excluded routine financial information, and required descriptions shorter than 180 characters. Empty results were allowed when no relevant announcement was found.

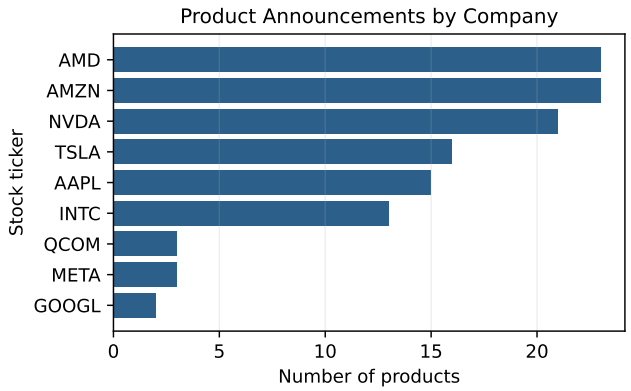


Figure 1: LLM-extracted product mentions by stock ticker.

3 Results

The pipeline completed processing of 100 filings and produced 119 product records across nine companies. No fields were missing, and all 119 descriptions were shorter than 180 characters. The largest numbers of extracted records came from Amazon and AMD (23 each), followed by NVIDIA (21).

Table 1: Summary of extraction output.

Measure	Value
Processed filings	100
Extracted product records	119
Companies with records	9
Filing groups with records*	29
Descriptions below 180 characters	119 / 119

*Estimated from unique ticker and filing-time pairs; the product CSV does not retain filing URLs.

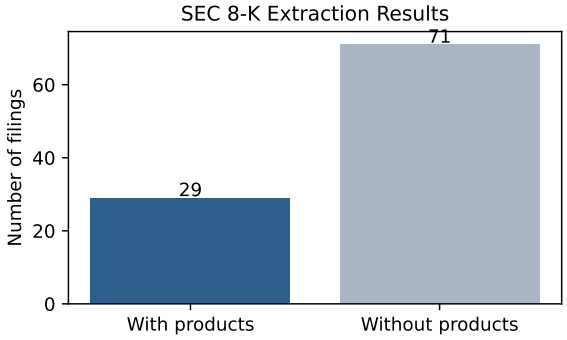


Figure 2: Estimated filings with and without extracted products.

4 Limitations and Conclusion

The model returned multiple product mentions from some filings, but these should not all be treated as verified launches. Some may describe existing products, planned features, or business collaborations. The current CSV also lacks document-level source URLs, limiting direct traceability. API rate limits temporarily interrupted processing and required retries.

Overall, the workflow demonstrates a practical way to structure product-related information from SEC filings. A useful next step is to retain each source URL and manually verify a sample of the extracted records.

GitHub: `https://github.com/DarinkaTownsend/LLM-Document-Analysis.git`